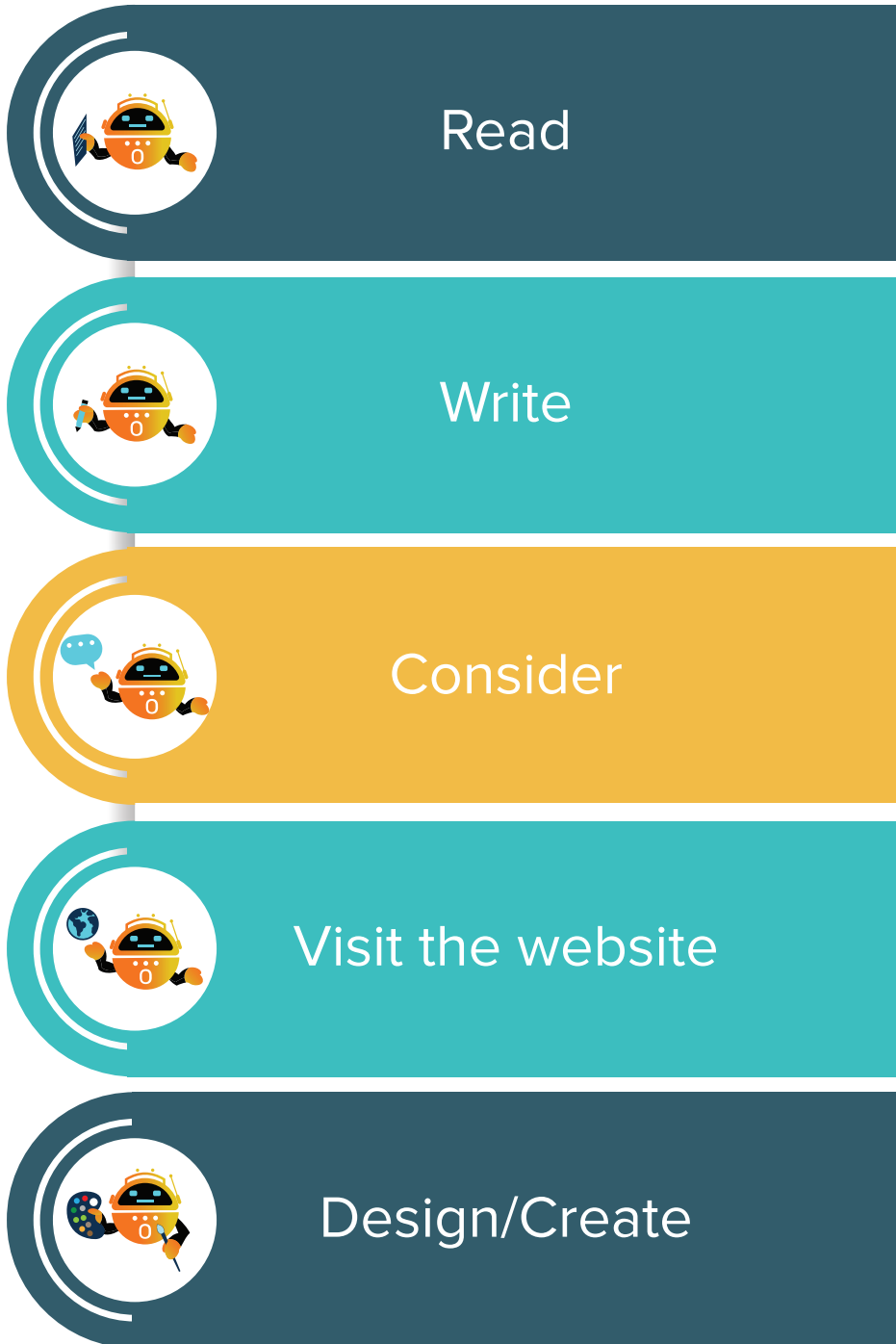




Designing and Building AI Products and Services Workbook

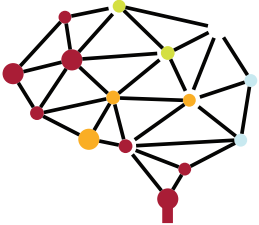


Your AI Mascot will guide you through this module, so watch out for these icons:





Workbook 1: Introduction to the Artificial Intelligence Design Process



Module 1

The key objective of this deliverable is to go through the four stages of AI product design and apply the learnings of this module to a proposed project of your choice. This can be an actual project that you want to undertake in your organization or a hypothetical scenario that you will use as an example to illustrate this module's concepts. This exercise is mainly intended to make you aware of all the items that need to be considered in the design process, so it is fine if you do not fully address all the points. If this is the case, make sure to include a list of questions that you would have to answer to figure out the details needed to address a particular item.

Your AI product design can be described by addressing each of the four-stage framework points presented in this module's lessons.

1

Stage One: Intelligence

1. Starting with the first stage, Intelligence, describe the choices you would make in regard to:

a. What is the scope and desired behaviors of your AI product?

Type your response here

Desired behavior:

- Improved voice recognition
- Contextual awareness for accurate responses

b. Identify the performance metrics you will use to measure the AI behavior.

Type your response here

Performance metrics:

1. Accuracy
2. Response time
3. User satisfaction
4. Task completion rate
5. Engagement

2

Stage Two: Business Process

2. Regarding the second stage, Business Process:

a. What strategic role will AI play in your long-term sustainable advantage?

Type your response here

Strategic role of AI: To provide personalized, proactive, and intelligent support for long-term user satisfaction and engagement.

b. What is the operational description of your AI product? In what business process will the AI intervene? What are some sensible improvement targets that you want your AI to achieve in terms of the performance metrics discussed above?

Type your response here

- Data collection and processing: Gather user data from calendars, emails, device usage, and locations.
- Natural language processing (NLP): Enhance the ability to understand and process natural language inputs.
- Predictive analytics: Utilize machine learning for predictions and continuous improvement.
- Personalization: Tailor responses and suggestions to individual user preferences.
- Security and privacy: Implement robust measures to protect user data.

3

Stage Three: AI technology

3. You will now consider the third stage: AI technology. Since this is a rather technical section, and you may not have all the relevant information without consulting with your team of specialists, it is fine if you hypothesize as long as you provide a rationale for your choices.

a. Pick a specific AI technology that you can incorporate in your product. What are its intellectual property implications in terms of challenges and opportunities?

Type your response here

Focus on personalization:

Challenges:

- Data privacy and ownership
- Patentability of AI algorithms
- Risk of copycat technologies

Opportunities:

- Patenting unique personalization techniques
- Trademarking brand elements
- Licensing opportunities

b. Choosing from one of the examples explored in the video lessons, what data approach will you use? Why is your chosen approach the best suited to your case?

Type your response here

Data approach: Inspired by Apple's emphasis on user privacy, using on-device processing and privacy-preserving techniques such as federated learning or differential privacy.

4

Stage Four: Tinkering

4. Finally, you will now explore the fourth stage: Tinkering.

- a. What key points will you need to consider in your software development plan? To the best of your knowledge, describe how you would implement each of them.*

Type your response here

Software development considerations:

- User privacy and data security: Compliance with regulations and robust security measures.
- Data collection and management: Efficient methods for data collection and processing.
- Natural language processing (NLP): Incorporate and optimize NLP algorithms.
- Machine learning and personalization algorithms: Continuous model updates based on user feedback.
- User interface and experience: Design an intuitive interface with various input methods.
- Integration with external services and devices: Seamless integration with services and IoT devices.
- Scalability and performance: Plan for scalability and optimize performance.
- Testing and quality assurance: Comprehensive testing for functionality, usability, security, and performance.
- Feedback and continuous improvement: Mechanisms for collecting user feedback and enhancing the experience.

b. Similar to what you did in the activity on AI cancers, address the following points, but this time as they apply to your AI product:

i. Identify one type of AI cancer that you can anticipate.

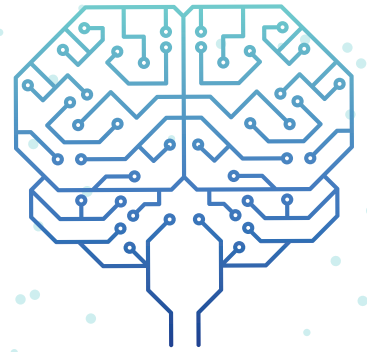
Type your response here

Anticipated AI challenge: Bias

ii. Discuss how you would prevent or mitigate the effects of the anticipated AI challenge.

Type your response here

- Ensure diversity in training data.
- Regularly monitor for bias.
- Implement fairness measures for equitable personalization across user groups.



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